

Econ 646 Spring 2013
Homework 4

This homework is due on March 27th, at the beginning of the lecture.

1. Consider the following problem:

$$\max_{\{C_i\}_{i=1}^m} \left[\sum_{i=1}^m C_i^{1-\frac{1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} \quad \text{subject to} \quad \sum_{i=1}^m C_i P_i = Z$$

Notice that this is the discrete version of the problem we discussed in the lecture. Show that the solution of this problem (i.e., the maximized value of the consumption index) is

$$C^* = \frac{Z}{P}$$

and the demand for individual good is

$$C_i = \left(\frac{P_i}{P} \right)^{-\varepsilon} C^*$$

where P is an appropriately defined price index, whose expression you have to find by yourself.